MATH11219

LECTURE NOTES

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Document Start Date: 27th of Jan 2026
Latest Update: February 18, 2026

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1 Differentiation

Differentiation (Derivatives) is just a mathematical expression of a functions rate-of-change. Think of the gradient of a function, area/estimation of a function, kinematics & optimisation of area or numbers.

1.1 Average Rate-of-Change (ARoC):

The ARoC of the function is just the estimation of a curve based on a linear lines gradient. I.E. the average gradient between two points.

$$\text{ARoC} = \frac{y_2 - y_1}{x_2 - x_1}$$

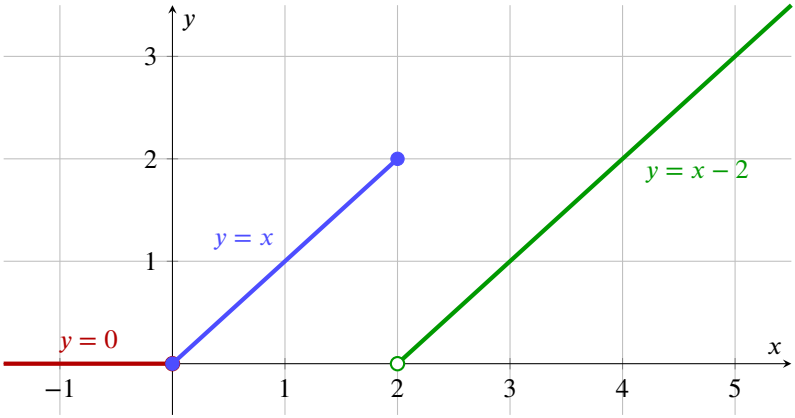
1.2 Instaneous Rate-of-Change (IRoC):

The ARoC of a function is good enough to understand the general direction in which a graph is continuing. Its accuracy can be improved by limiting the amount of change between x_1 & x_2. However a problem occurs when $\Delta x \to 0$. As cannot divide by 0. To find a functions IRoC, use the derivatives by first principles.

$$\begin{aligned}
f(x) &= 3x^2 + 2 \\
f'(x) &= \lim_{\delta x \to 0} \frac{f(x + \delta x) - f(x)}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3(x + \delta x)^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3x^2 + 6x\delta x + 3\delta x^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{6x\delta x + 3\delta x^2}{\delta x} \\
&= \lim_{\delta x \to 0} 6x + 3\delta x \\
&= 6x
\end{aligned}$$

1.3 Limits in Functions:

Specifically in Piece-wise functions there can be an issue with derivatives and their interpretations. This is mostly around when the Left and Right Hand Limits not meeting on the same point, making the functions discontinuous.



As can be seem in the graph, at the point $x = 2$ there is not derivative as the function is discontinuous. However there moving in, up until that point a derivative is present. A similar story is true at the point $x = 0$. All other points along the x-axis have a derivative that exists.

1.4 Common Derivatives:

$y = f(x)$	$\frac{dy}{dx}$
c	0
x^n	$n \cdot x^{n-1}$
e^{kx}	$k \cdot e^{kx}$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{\ln(a) \cdot x}$
$\ln(x)$	$\frac{1}{x}$
$\sin(ax+b)$	$a\cos(ax+b)$
$\cos(ax+b)$	$-a\sin(ax+b)$
$\tan(ax+b)$	$a\sec^2(ax+b)$
$\csc(ax+b)$	$-a\csc(ax+b)\cot(ax+b)$
$\sec(ax+b)$	$a\sec(ax+b)\tan(ax+b)$
$\cot(ax+b)$	$-a\csc^2(ax+b)$
$\arcsin(ax+b)$	$\frac{a}{\sqrt{1-(ax+b)^2}}$
$\arccos(ax+b)$	$\frac{-a}{\sqrt{1-(ax+b)^2}}$
$\arctan(ax+b)$	$\frac{a}{1+(ax+b)^2}$
$\sinh(ax+b)$	$a\cosh(ax+b)$
$\cosh(ax+b)$	$a\sinh(ax+b)$
$\tanh(ax+b)$	$a\text{sech}^2(ax+b)$
$\text{csch}(ax+b)$	$-a\text{csch}(ax+b)\coth(ax+b)$
$\text{sech}(ax+b)$	$-a\text{sech}(ax+b)\tanh(ax+b)$
$\coth(ax+b)$	$-a\text{csch}^2(ax+b)$
$\text{arcsinh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2+1}}$
$\text{arccosh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2-1}}$
$\text{arctanh}(ax+b)$	$\frac{a}{1-(ax+b)^2}$

2 Techniques of Differentiation

2.1 Product Rule:

The product rule is used to differentiate two functions multiplied together.

$$\begin{aligned} y &= f(x)g(x) \\ \frac{dy}{dx} &= f'(x)g(x) + f(x)g'(x) \end{aligned}$$

$$\begin{aligned} y &= uv \\ \frac{dy}{dx} &= u'v + v'u \end{aligned}$$

2.2 Quotient Rule:

This rule is used when two functions divide each other. It is important to know that the order of functions matter with this rule. As this subtracts rather then addition.

$$\begin{aligned} y &= \frac{u}{v} \\ \frac{dy}{dx} &= \frac{u'v - v'u}{v^2} \end{aligned}$$

2.3 Chain Rule:

By using differentiation by substitution (A.K.A) chain rule, function within other functions can be derived.

$$\begin{aligned} y &= f(g(x)) \\ \frac{dy}{dx} &= g'(x)f'(g(x)) \end{aligned}$$

2.4 Implicit Differentiation

This is used when the equation is not simply $y = f(x)$, e.g.:

$$x^2 - y^3 + \sin(x) - \cos(y) = 1$$

$$\begin{aligned}
\frac{d}{dx}(x^2) &= 2x \\
\frac{d}{dx}(-y^3) &= -3y^2\frac{dy}{dx} \\
\frac{d}{dx}(\sin(x)) &= \cos(x) \\
\frac{d}{dx}(-\cos(y)) &= \sin(y)\frac{dy}{dx} \\
\frac{d}{dx}(1) &= 0
\end{aligned}$$

$$\begin{aligned}
\frac{d}{dx}(x^2 - y^3 + \sin(x) - \cos(y)) &= \frac{d}{dx}1 \\
2x - 3y^2\frac{dy}{dx} + \cos(x) + \sin(y)\frac{dy}{dx} &= 0 \\
\frac{dy}{dx} &= \frac{-2x - \cos(x)}{\sin(y) - 3y^2} \\
&= \frac{2x + \cos(x)}{3y^2 - \sin(y)}
\end{aligned}$$

2.5 Parametric Differentiation

Basically just multiplying the derivatives together.

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

2.6 Logarithmic Differentiation

Just follow the basic steps and the derivative will be found. Mostly useful for a short version of the product rule.

1. Find the log of each side of the equation, & use log laws to expand the equation
2. Differentiate each side with respect too the independent variable
3. Rearrange and substitute
4. Simplify

For Example,

$$y = t^2(1+t)^3$$

$$\begin{aligned}\ln(y) &= \ln(t^2[1+t]^3) \\ &= \ln(t^2) + \ln([1+t]^3) \\ &= 2\ln(t) + 3\ln(1+t)\end{aligned}$$

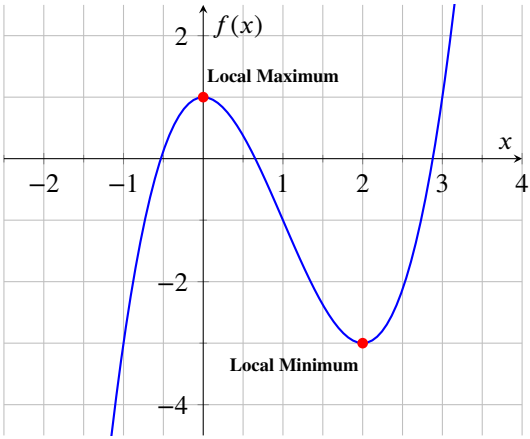
$$\begin{aligned}\frac{d}{dt}\ln(y) &= \frac{d}{dt}[2\ln(t) + 3\ln(1+t)] \\ \frac{1}{y}\frac{dy}{dt} &= \frac{2}{t} + \frac{3}{1+t} \\ \frac{dy}{dt} &= y(\frac{2}{t} + \frac{3}{1+t})\end{aligned}$$

2.7 Higher Derivatives

Found by differentiating the function multiple times.
These are useful as they can describe the nature of a function, such as
If $\frac{dy}{dx} > 0$, the gradient can be said to be increasing, or $\frac{dy}{dx} < 0$, the gradient is decreasing and $\frac{dy}{dx} = 0$ the gradient is stable.
Or $\frac{d^2y}{dx^2} > 0$, concavity is up and the point (x,y) is described as a minimum. $\frac{d^2y}{dx^2} = 0$, described as a Point of Inflection. Basically just the point where the concavity of a function changes. $\frac{d^2y}{dx^2} < 0$, concave down and point (x,y) is a maximum.

3 Applications of Derivatives

3.1 Maximum & Minimum Points



These points are found by substituting the x-values found when $\frac{dy}{dx} = 0$ into $\frac{d^2y}{dx^2}$. In addition POIs can be found when $\frac{d^2y}{dx^2} = 0$. A simple way for remembering which point is what is understanding how the gradient is achieved around the point.

3.2 Points of Inflextion (POI)

This is the point in which the graph changes in concavity.
Maximum - Concave Down
Minimum - Concave Up

3.3 Newton-Raphson Method

This is an estimation method used to find roots within an equation. The reason for this is for polynomials not able to be found by hand. For the method to work it assumes point x_2 is closer too the root value than x_1. However the method can come off course as it is entirely dependent upon the intial value chosen to represent point x_1. As this can cause the approximation to diverge (rather than converge) upon a particuar point.

$$x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$

4 Sequences & Series:

4.1 Sequences

A sequence is a set of numbers in a particular order with most cases following a pattern.
E.g. $x[k] = (-1)^{k+1} \cdot \frac{1}{k},\ \forall k \geq 1$ becomes,
$1, \frac{-1}{2}, \frac{1}{3}, \ldots$

4.2 Series

Series are the sum of multiple sequences up until a certain point, eg.

$$\sum_{n=1}^{5}(2n-1) = 1+3+5+7+9$$

4.3 Convergence

If an infinite series is said to converge upon a value if and only if it tends towards a singular value. For example $\lim_{k\to\infty}\ \frac{1}{k}$ converges towards 0. A sequence can converge on a non-zero quantity. For a point that might be considered to converge on ∞ is instead called to diverge. E.g. $\lim_{k\to\infty} \sqrt{k} = \infty$.

$$x[k] = \frac{3k^2+2k-7}{9k^2-7k},\ k \to \infty$$

$$\begin{aligned}
&\lim_{k\to\infty} \frac{3k^2+2k-7}{9k^2-7k} \\
=&\lim_{k\to\infty} \frac{(3k^2+2k-7)\times\frac{1}{k^2}}{(9k^2-7k)\times\frac{1}{k^2}} \\
=&\lim_{k\to\infty} \frac{3+\frac{2}{k}-\frac{7}{k^2}}{9-\frac{7}{k}} \\
=&\frac{3}{9}\text{ , because } \lim_{k\to\infty}\frac{a}{k} = 0 \\
=&\frac{1}{3}
\end{aligned}$$

4.4 Arithmetic Sequences

This is a type of sequence in which each values difference is the same amount.

$$a_n = a_1 + (n-1)d$$

These vaues can also be summed to a total series. The proof of this is beautiful thing to see, look it up online.

$$S_n = \frac{n}{2}(2a_1 + [n-1]d)$$

4.5 Geometric Sequences

This a sequence of numbers related to a common ratio of r.

$$a_n = a_1 \cdot r^{n-1}$$

$$S_n = \frac{a_1(1-r)}{1-r},\ r \neq 1$$

If $-1 \leq r \leq 1$,

$$S_n = \frac{a_1}{1-r}$$

This is also the only one that can find S_∞. Outside of $-1 \leq r \leq 1$ r^n fails to converge on a finite limit as $n \to \infty$, thus the series and sequence diverges.

4.6 Power Series

$$\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\
\sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \approx x \\
\cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \approx 1 - \frac{x^2}{2!} \\
\log(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots,\ -1 \leq x \leq 1
\end{aligned}$$

4.7 Iterative Solutions to Non-linear Equations

$$\begin{aligned}
f(x) &= e^{-x} - x = 0 \\
\therefore x &= e^{-x} \\
x_{n+1} &= e^{x_n}
\end{aligned}$$

5 Taylor & Maclaurin Series

5.1 Approximating Points on a Function

Think amount the point (Q) on a function. When an approximation of the function is close to Q it is considered to be fairly accurate. However as the distance between the approximation and Q grows.

$$f(x) = f(a) + f'(a)(x - a)$$

5.2 Taylor Series

Taylor Series becomes more accurate the higher order (highest power in the set) is.
Zero: $p_0(x) = f(a)$
First: $p_1(x) = f(a) + f'(a)(x - a)$
Second: $p_2(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2$
Continues with higher orders

5.3 Remainder Term

This is the error in the nth term of a Taylor Polynomial.

$$f(x) = p_n(x) + R_n(x)$$

$$R_n(x) = \frac{y^{n+1}(c)(x - a)}{(n + 1)!}, \text{ where } a < c < x$$

5.4 Maclaurin Series

Think of this as a Taylor polynomial, except that it rotates around the point $x = 0$.

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!} + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$

6 Intergration

This is just anti-differentiation.

6.1 Common Derivatives

$f(x)$	$F(x)$
k	$kx + c$
x^n	$\frac{1}{n+1}x^{n+1} + c$
x^{-1}	$\ln(x) + c$
e^{ax}	$\frac{1}{a}e^{ax} + c$
$\sin(ax+b)$	$-\frac{1}{a}\cos(ax+b) + c$
$\cos(ax+b)$	$\frac{1}{a}\sin(ax+b) + c$
$\tan(ax+b)$	$\frac{1}{a}\ln[\sec(ax+b)] + c$
$\csc(ax+b)$	$\frac{1}{a}\ln[\csc(ax+b) - \cot(ax+b)] + c$
$\sec(ax+b)$	$\frac{1}{a}\ln[\sec(ax+b) + \tan(ax+b)] + c$
$\cot(ax+b)$	$\frac{1}{a}\ln[\sin(ax+b)] + c$
$\frac{1}{\sqrt{a^2-x^2}}$	$\arcsin(\frac{x}{a}) + c$
$\frac{-1}{\sqrt{a^2-x^2}}$	$\arccos(\frac{x}{a}) + c$
$\frac{1}{a^2+x^2}$	$\frac{1}{a}\arctan(\frac{x}{a}) + c$

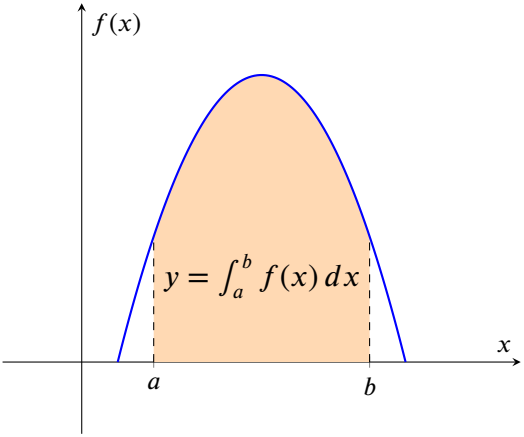
6.2 Kinematics

Kinematics is just the algebra of motion.
Distance $\rightarrow$ x
Velocity $\rightarrow$ $v = \frac{dy}{dx}$
Accelaration $\rightarrow$ $a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = v\frac{dv}{dx} = \frac{d}{dx}(\frac{1}{2}v^2)$

6.3 Trig Functions

Trig functions presented to a power (e.g. $\sin^5(x)$) cannot be easily integrated. As such, for these to be integrated, they must come down to an index of 1, through the use of trigonometric identities to seperate out the powers.

6.4 Definite Integration



A note on definite integration. Make sure that the everything between the boundaries is either above or below the relevant axs. Else multiple integration will be required as some of the area will be considered negative.

7 Techniques of Integration

7.1 Integration by Parts:

When thinking about the product rule ($\frac{d}{dx}(uv) = u'v + v'u$), integration by parts is a way of doing a reverse product rule.

$$\int u'v = uv - \int uv'$$

The key to solving the above equations is making smart choices about which parts of the functions are u and v. The following guide can be utilized when making that descision:

1. Logs
2. Inverse Trigonometric
3. Algebraic (Polynomials)
4. Normal Trigonometric
5. Exponential

For Example:

$$\begin{aligned}
&\int (5x+1)\cos(2x)dx \\
&\text{let } u = 5x+1 \quad v' = cos(2x) \\
&\qquad u' = 5 \qquad v = \frac{1}{2}\sin(2x) \\
&\therefore \int uv' = uv - \int u'v \\
&\qquad = \frac{1}{2}(5x+1)\sin(2x) - \int \frac{5}{2}\sin(2x)dx \\
&\qquad = \frac{1}{2}(5x+1)\sin(2x) + \frac{5}{4}\cos(2x) + c
\end{aligned}$$

7.2 Integration via Substitution:

This allows a chain rule derivative to be integrated. It does this by splitting the terms down into usable terms. However, it is important to note that the seeminly obvious solution is not always the correct one. E.g.

$$\int x\sqrt{1-x^2}dx$$

$$\begin{aligned} \text{let } u &= 1-x^2 \\ \frac{du}{dx} &= -2x \\ dx &= -\frac{du}{2x} \end{aligned}$$

$$\begin{aligned} \int x\sqrt{1-x^2}dx &= \int x\sqrt{u}(-\frac{du}{2x}) \\ &= -\frac{1}{2}\int \sqrt{u}du \\ &= -\frac{1}{2}\cdot\frac{2}{3}u^{\frac{3}{2}} + c \\ &= -\frac{1}{3}(1-x^2)^{\frac{3}{2}} + c \end{aligned}$$

7.3 Integrals in Forms:

When $\int \frac{f'(x)}{f(x)} = \ln \|f(x)\| + c$. This can just be a simple way of immediately recognizing an integration via substitution.

7.4 Integration via Partial Fractions:

Turn a integral not in form (i.e. not substitution) into standard natural log form. This can then be integrated as normal.

8 Further Integration

8.1 Average Value of a Function:

Often wanting to know the average value within a function is useful. The independent variable within mostly a period of time over a specified interval. The main example of this is time. I.E. finding the average velocity of car going certain speeeds.

$$\overline{x} = \frac{1}{b-a}\int_a^b f(x)dx$$

E.g. Find the Average value of $f(t) = t^2$, $t \in [1, 3]$

$$\begin{aligned} AV =& \frac{1}{3-1}\int_1^3 t^2 dt \\ =& \frac{1}{2}[\frac{1}{3}t^3]_1^3 \\ =& \frac{1}{2}([\frac{27}{3}] - [\frac{1}{3}]) \\ =& \frac{1}{2}(\frac{26}{3}) \\ =& \frac{13}{3} \end{aligned}$$

8.2 RMS Value of a Function:

The RMS value of a function is the 2nd root of the mean value over a given period.

$$RMS = \sqrt{\frac{1}{b-a}\int_a^b [f(t)]^2 dt}$$

One of the main applications for this in alternating currents & voltages between DC & AC circuits.

8.3 Volume of Revolution:

Rotating a function (here $f(x)$) around an axis (x), producing a 3D shape. Most of the time makign a cone-like shape.

$$V = \pi\int_a^b [f(x)]^2 dx$$

E.g. Find the volume of $y = 2x$ for $0 \leq x \leq 3$

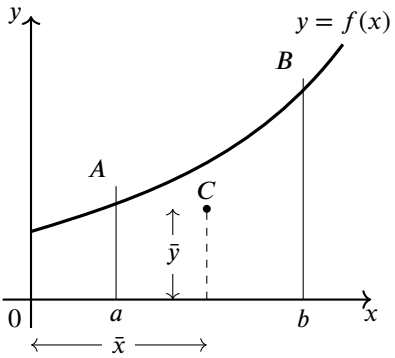
$$\begin{aligned} V =& \pi \int_0^3 [2x]^2 dx \\ =& \pi \int_0^3 4x^2 dx \\ =& 4\pi [\frac{1}{3}x^3]_0^3 \\ =& 4\pi([\frac{27}{3}] - [0]) \\ =& 36\pi \text{units}^3 \end{aligned}$$

8.4 Centriod of a Plane Figure:

When looking for the point at which the whole acts. This means that the overall mass could be replaced with a point at this location (E.g. Center of Gravity). It could then be said that this is the balance point for the plane.

$$Area\ x \approx \sum_{x=a}^{x=b} xy\Delta x \qquad \overline{x} = \frac{\int_a^b xydx}{\int_a^b ydx}$$

$$Area\ y \approx \sum_{x=a}^{x=b} \frac{1}{2}y^2\Delta x \qquad \overline{x} = \frac{\int_a^b y^2 dx}{2\int_a^b ydx}$$



8.5 Center of Gravity of a Solid Revolution:

$$\overline{x} = \frac{\int_{x=a}^{x=b} xy^2 dx}{\int_{x=a}^{x=b} y^2 dx}$$
$$\overline{y} = 0$$

The above formulas assume a rotation around the x-axis. Swap $x \to y$ for rotations around the y-axis.

8.6 Length of a Curve:

Measures the actual length of the arc of a curve between the two points.

$$s = \int_a^b \sqrt{1 + (\frac{dy}{dx})^2} dx$$

8.7 Surface Area of Revolution:

$$SA = 2\pi \int_a^b y\sqrt{1 + (\frac{dy}{dx})^2} dx$$

8.8 Volumes of Washers:

8.8.1 Basic Formula

This only works when rotating around an axis

$$V = \pi \int_a^b [f(x)]^2 - [g(x)]^2 dx$$

8.8.2 Complex Formula

When rotating around a specific point. A note with these is that the second function is only used when a second function is also being used.

About $y = c$:

$$V = \pi \int_{x=a}^{x=b} [c - f(x)]^2 - [c - g(x)]^2 dx$$

9 Ordinary Differentiation

9.1 Numerical Integration:

Numerical Integration is a set of principles to follow for equations that cannot be solved analytically (I.E. $\sin(x^2)$). The two most common techniques to are approximation of the solution to the integration of the formula. These are the Trapezoidal and Simpson's Rule.

9.1.1 Trapezoidal Rule

$$\int_a^b f(x)dx \approx \frac{b-a}{2n}[f(x_0) + 2(f(x_1) + f(x_2) + \ldots + f(x_{n-1})) + f(x_n)]$$

9.1.2 Simpson's Rule

Similar too the Trapezoidal Rules, Simpson's Rule provides an approximation that is considered even more accurate at width h ($\frac{b-a}{n}$). It does this through utilizing the sum of multiqle quadratic equations.

$$\int_a^b f(x)dx \approx \frac{h}{3}[f(x_0) + 4[f(x_{\text{odd}})] + 2[f(x_{\text{even}})] + f(x_n)]$$

9.2 Standard Ordinary Differential Equations:

$$\begin{aligned} F_1 = -ky \quad F_2 &= ma \\ F_1 &= F_2 \\ \therefore -ky &= ma \\ a &= -\frac{ky}{m} \\ \frac{d^2y}{dx^2} &= -\frac{ky}{m} \\ m\frac{d^2y}{dx^2} + ky &= 0 \end{aligned}$$

Above is an example of a second-order differential equation.

9.3 Linear Differential Equations:

A differential equation is said to be linear, if and only if:

1. The dependent variable and derivative are of the first order (index 1)
2. No products of dependent variable and derivative are present
3. No non-linear functions are present within the equation

9.4 Seperation of Variables:

$$\frac{dy}{dx} = f(x)g(y)$$
$$\int \frac{1}{g(y)} dy = \int f(x)dx$$

9.5 Standard Form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$
$$y(x) = \frac{\int \mu Q(x)dx}{\mu}, \text{ where } \mu = e^{\int P(x)dx}$$

10 Ordinary Differential Equations

10.1 Second Order Linear Equations:

$$p(x)\frac{d^2y}{dx^2} + q(x)\frac{dy}{dx} + r(x)y = f(x)$$

This is the standard form of second order linear equations. Where each function can be of x only. The solution to this integral is $y_p(x)$ (A.K.A the particular integral).
The general solution is $y(x) = y_H(x) + y_p(x)$.

10.1.1 Homogenous equations

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x) \text{ or } 0$$
$$ak^2 + bk + c = 0$$

Type of Root	Roots Equality	General Solution
Unique	$b^2 - 4ac > 0,\ k_1 \neq k_2$	$y_H(x) = Ae^{k_1 x} + Be^{k_2 x}$
Repeated	$b^2 - 4ac = 0,\ k_1 = k_2$	$y_H(x) = Ae^{kx} + Bxe^{kx}$
Complex	$b^2 - 4ac < 0,\ k_{1,2} = \alpha \pm \beta j$	$y_H(x) = e^{\alpha x}[A\cos(\beta x) + B\sin(\beta x)]$

10.2 Non-homogenous Constant Coefficients:

Form of $f(x)$	Form of Integral
Constant	$y_p(x) = c$
Polynomial of Degree m	$y_p(x) = Ax^m + Bx^{m-1} + \ldots + Yx + Z$
$\cos(mx)$ and/or $\sin(mx)$	$y_p(x) = A\cos(mx) + B\sin(mx)$
ae^{mx}	$y_p(x) = Ae^{mx}$
$e^{mx}\cos(mx)$ and/or $e^{mx}\sin(mx)$	$y_p(x) = [A\cos(mx) + B\sin(mx)]$

To find the specific values of A and B in the table above, substitute the derivatives into the original ordinary differential equation.